

Discrete structure for computing

Assignment #2

1. Let p , q , and r be the propositions

p : You have the flu.

q : You miss the final examination

r : You pass the course

Express each of these propositions as an English sentence.

$$(p \wedge q) \vee (\neg q \wedge r)$$

2. Let p , q , and r be the propositions

p : You get an A on the final exam.

q : You do every exercise in this book.

r : You get an A in this class.

Write these propositions using p , q , and r and logical connectives (including negations).

You get an A in this class if and only if you either do every exercise in this book or you get an A on the final.

3. State the converse, contrapositive and inverse of each of these conditional statements.

c) When I stay up late, it is necessary that I sleep until noon

4. Construct a truth table for each of these compound propositions.

$$d) (p \wedge q) \rightarrow (p \vee q)$$

5. Construct a truth table for each of these compound propositions.

$$e) (p \vee q) \wedge \neg r \quad f) (p \wedge q) \vee \neg r$$

6. Use truth tables to verify the associative laws

$$b) (p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

7. Show that $(p \rightarrow q) \vee (p \rightarrow r)$ and $p \rightarrow (q \vee r)$ are logically equivalent

8. Show that the compound proposition below is a contradiction:

Via truth tables; Via equivalences:

$$(p \vee q) \wedge (\neg p \vee q) \wedge (p \vee \neg q) \wedge (\neg p \vee \neg q)$$

9. You go to your digital circuit lab to implement a boolean function represented by the following com

$$(p \wedge q) \vee (\neg q \wedge p) \vee (r \wedge p) \vee (q \wedge r).$$

10. Use the Euclidean algorithm to calculate $\gcd(100, 270)$. Show each step.

11. Let f and g be the functions from the set of integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition of f and g ? What is the composition of g and f ?

12. Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} - a_{n-2}$ for $n = 2, 3, 4, \dots$, and suppose that $a_0 = 3$ and $a_1 = 5$. What are a_2 and a_3 ?

13. The Fibonacci sequence, $f_0, f_1, f_2, \dots$, is defined by the initial conditions $f_0 = 0, f_1 = 1$, and the recurrence relation $f_n = f_{n-1} + f_{n-2}$

for $n = 2, 3, 4, \dots$.

Find the Fibonacci numbers f_2, f_3, f_4, f_5 , and f_6 .

14. Find the solution to each of these recurrence relations with the given initial conditions. Use an iterative approach

b) $a_n = a_{n-1} + 3, a_0 = 1$

c) $a_n = a_{n-1} - n, a_0 = 4$

d) $a_n = 2a_{n-1} - 3, a_0 = -1$

e) $a_n = (n + 1)a_{n-1}, a_0 = 2$